Instant Recovery Monitoring System – Documentation

❖ Project Overview

The **Instant Recovery Monitoring System** is a workflow-based solution built using **Microsoft Power Apps**, **Power Automate**, and **SharePoint Online**. It automates the process of recording, tracking, and notifying recovery test results for virtual machines (VMs) within the IT infrastructure.

This system ensures that all recovery test results are logged accurately, automatically notifies relevant managers, and stores all submissions for auditing and reporting purposes.

❖ Objectives

- Automate the monitoring and reporting process of Instant Recovery operations.
- Ensure accurate tracking of backup, restore, and issue details through SharePoint integration.
- Enable IT members to record and submit recovery data easily via Power Apps.
- Notify Managers automatically based on status
- Generate consistent formatted emails with color-coded status indicators
- Track the IT member that submit the status automatically

❖ Tools & Technologies Used

- **Microsoft Power Apps**
- **Microsoft Power Automate**
- **SharePoint**
- **Microsoft Outlook (Email Integration)**

◆ System Components

1. SharePoint List – “Instant Recovery Log”

This list serves as the **central data repository** for all recovery entries.

Columns:

Field Name	Type	Description
Date	Date/Time	The date entry is created (auto-filled by the user during entry).
IT Member	Person or Group	Auto-filled with the logged-in user’s name (captured from Power Apps).
VM Name	Single line of text	Name of the virtual machine tested for instant recovery.
Backup Date	Date/Time	The date of the latest backup used in the recovery test.
Restore Test Date	Date/Time	The date the restore test was performed.
Status	Choice	Indicates the outcome of the recovery test. (Successful / Warning / Failed)
Comment	Multiple lines of text	Additional notes or observations by the IT member.
Attachments	Attachments	Allows attaching screenshots or reports related to the test.

2. Power Apps – Data Entry Form

A custom Power App form was designed and linked to the SharePoint list.

Main Features:

- Automatically fills **IT Member** field using:

  [User().FullName]

and stores it in the person column.

- Provides dropdowns for **Status** (connected to SharePoint choice field).
- Date pickers for **Backup Date**, **Restore Test Date**, and **Date**.

- Text inputs for **VM Name** and **Comments**.
- File attachment control for evidence upload.
- Submits the form directly to the SharePoint list using:

  [SubmitForm(Form1)]

3. Power Automate – Automated Notification Flow

Once a new item is created in the SharePoint list, the **Power Automate flow** triggers automatically.

Trigger:

When an item is created in “Instant Recovery Log”.

Actions in Flow:

- **Condition Check:**
  - Reads the value of the “Status” column.
  - If **Status = Successful** → send success email.
  - If **Status = Warning** or **Failed** → send alert email (different color/style).

- **Email Formatting:**
  - Email body is built using **HTML table** formatted similarly to your provided template (with green or red highlights).
  - Example (Successful):

    ```
    <p>Dears,</p>
    <p>Instant Recovery has been completed
    <b><strong
    style="color:#2e7d32;">Successfully</strong></b>
    for the following:</p>
    ```

  - For **Failed**, the color changes to **#d32f2f (red)**.
  - For **Warning**, the color changes to **# ff8f00 (orange)**.

- **Dynamic Content Mapping:**
  - Each column is dynamically inserted using:

    @{triggerBody()?['VM_x0020_Name']}
    @{triggerBody()?['Backup_x0020_Date']}
    @{triggerBody()?['Restore_x0020_Test_x0020_Date']}

  - Includes the “Comment” and “IT Member”.
- **Email Recipients:**
  - **To:**
    - IT Infrastructure Manager
    - Service Desk Manager
  - **CC:**
    - ITSupport3
    - HOSupport
- **Subject:**
  - For Success:

✅ Instant Recovery Test Completed Successfully – @{triggerBody()?['VM_x0020_Name']}

  - For Warning/Failed:

❌ Instant Recovery Test Warning/Failed – @{triggerBody()?['VM_x0020_Name']}

⚙️ Power Automate Workflow

Flow Name: Instant Recovery Notification

🔸 Trigger

- **Type:** When an item is created or modified
- **Source:** SharePoint list Instant Recovery

◆ Logic (Switch Control)

Switch is used to check the **Status** field value and format the email accordingly.

Example Expression:

```
switch(
  triggerBody()?['Status/Value'],
  'Successfully', 'Green',
  'Failed', 'Red',
  'Completed with Warning', 'Orange'
)
```

◆ Email Body

HTML-based email with color-coded status inside a styled table

```
<table style="border-collapse:collapse; width:100%; border:1px solid #ccc;">
  <tr style="background-color:#f2f2f2;">
    <th style="padding:8px;">VM Name</th>
    <th style="padding:8px;">Date</th>
    <th style="padding:8px;">IT Member</th>
    <th style="padding:8px;">Status</th>
  </tr>
  <tr>
    <td style="padding:8px;">@{triggerBody()?['VMName']}</td>
    <td style="padding:8px;">@{triggerBody()?['Date']}</td>
    <td style="padding:8px;">@{triggerBody()?['ITMember']?['DisplayName']}</td>
    <td style="padding:8px; color:@{variables('Color')}">
      @{triggerBody()?['Status']?['Value']}
    </td>
  </tr>
</table>
```

✉️ Email Notification Logic

Status	Email Color	Description
Successfully	Green	Completed successfully
Failed	Red	Has been failed
Completed with Warning	Orange	Completed with warning

🔹 System Workflow Summary

Step	Action	Tool Used
1	IT Support logs into Power Apps and fills the recovery form	**Power Apps**
2	Data submitted and saved to SharePoint	**SharePoint List**
3	Power Automate detects new entry	**Power Automate**
4	Flow builds an HTML email with recovery details	**Power Automate**
5	Email sent to managers with IT Support team in CC	**Power Automate**
6	Records stored for auditing and follow-up	**SharePoint**

🔹 Future Enhancements (Optional Ideas)

- Create a Power BI dashboard for visual tracking of recovery results.
- Add history logs for editing the existing case.
- Implement approval workflow for escalations.